

INSERT, UPDATE & DELETE

SQL Data Modification Commands – Practical Software Testing Guide

Commands covered: INSERT, UPDATE, DELETE, SELECT, WHERE, and NOW(). Includes realistic music, food-delivery, shopping, and login-module examples.

1. INSERT – Add 3 Playlists

```
CREATE TABLE playlist (  
    playlist_id INT PRIMARY KEY,  
    user_id INT,  
    playlist_name VARCHAR(100),  
    created_at DATETIME  
);  
  
INSERT INTO playlist  
    (playlist_id, user_id, playlist_name, created_at)  
VALUES  
    (101, 1, 'Morning Vibes', '2026-09-20 08:30:00'),  
    (102, 2, 'Workout Songs', '2026-09-21 07:15:00'),  
    (103, 3, 'Relaxing Music', '2026-09-22 21:00:00');  
  
SELECT * FROM playlist;
```

Example: A music app lets a user create a playlist. INSERT stores the new playlist records in the database. SELECT is then used to verify that the records were saved.

playlist_id	user_id	playlist_name	created_at
101	1	Morning Vibes	2026-09-20 08:30:00
102	2	Workout Songs	2026-09-21 07:15:00
103	3	Relaxing Music	2026-09-22 21:00:00

2. UPDATE – Edit the foodielover Profile

```
CREATE TABLE user_profile (  
    user_id INT PRIMARY KEY,  
    username VARCHAR(100),  
    email VARCHAR(150),  
    phone VARCHAR(20)  
);  
  
UPDATE user_profile  
SET  
    email = 'foodielover@gmail.com',  
    phone = '9988776655'  
WHERE username = 'foodielover';  
  
SELECT *  
FROM user_profile  
WHERE username = 'foodielover';
```

Example: This simulates a Zomato-style profile edit. UPDATE changes the email and phone only for the user named foodielover. The WHERE condition is important because it prevents updating every user's record.

user_id	username	email	phone
1	foodielover	foodielover@gmail.com	9988776655

3. DELETE – Remove Expired Sessions

```
CREATE TABLE sessions (  
    session_id INT PRIMARY KEY,  
    user_id INT,
```

```

    expiry_time DATETIME
);

DELETE FROM sessions
WHERE expiry_time < NOW();

SELECT * FROM sessions;

```

Example: This simulates expired login-session cleanup in an application such as a payment or shopping app. NOW() returns the current MySQL date/time. Any session whose expiry_time is earlier than the current time is deleted.

Condition	Meaning
expiry_time < NOW()	Session has already expired and can be removed

4. INSERT – Login Attempt Test Data

```

CREATE TABLE login_attempts (
    attempt_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    attempt_time DATETIME,
    status VARCHAR(20)
);

INSERT INTO login_attempts
    (user_id, attempt_time, status)
VALUES
    (1, '2026-09-22 09:00:00', 'success'),
    (1, '2026-09-22 09:05:00', 'failure'),
    (2, '2026-09-22 10:15:00', 'failure'),
    (2, '2026-09-22 10:20:00', 'success');

SELECT * FROM login_attempts;

```

Example: This creates realistic test data for a login module. Two users have different timestamps and both success and failure results, allowing a tester to validate login-history and security-related functionality.

attempt_id	user_id	attempt_time	status
1	1	2026-09-22 09:00:00	success
2	1	2026-09-22 09:05:00	failure
3	2	2026-09-22 10:15:00	failure
4	2	2026-09-22 10:20:00	success

5. UPDATE – Rename a Playlist and Verify

```

-- Check the existing playlist
SELECT *
FROM playlist
WHERE playlist_id = 101;

-- Update the playlist name
UPDATE playlist
SET playlist_name = 'Morning Energy'
WHERE playlist_id = 101;

-- Verify the change
SELECT *
FROM playlist
WHERE playlist_id = 101;

```

Example: The playlist named Morning Vibes is renamed to Morning Energy. The final SELECT verifies that the database contains the expected updated value.

playlist_id	user_id	playlist_name
101	1	Morning Energy

SQL Testing Quick Notes

Command	Purpose	Tester Example
INSERT	Add records	Create playlist / login test data
UPDATE	Modify records	Edit email, phone, playlist name
DELETE	Remove records	Delete expired sessions
SELECT	View and verify data	Confirm expected database state
WHERE	Target specific rows	Update only foodielover
NOW()	Current date/time	Identify expired sessions

Golden Rule for Database Testing: After INSERT, UPDATE, or DELETE, use SELECT to verify the actual database state and compare it with the expected result.

Complete Practice Script

```
-- INSERT playlists
INSERT INTO playlist
  (playlist_id, user_id, playlist_name, created_at)
VALUES
  (101, 1, 'Morning Vibes', '2026-09-20 08:30:00'),
  (102, 2, 'Workout Songs', '2026-09-21 07:15:00'),
  (103, 3, 'Relaxing Music', '2026-09-22 21:00:00');

-- UPDATE profile
UPDATE user_profile
SET email = 'foodielover@gmail.com',
    phone = '9988776655'
WHERE username = 'foodielover';

-- DELETE expired sessions
DELETE FROM sessions
WHERE expiry_time < NOW();

-- INSERT login attempts
INSERT INTO login_attempts
  (user_id, attempt_time, status)
VALUES
  (1, '2026-09-22 09:00:00', 'success'),
  (1, '2026-09-22 09:05:00', 'failure'),
  (2, '2026-09-22 10:15:00', 'failure'),
  (2, '2026-09-22 10:20:00', 'success');

-- UPDATE playlist and verify
UPDATE playlist
SET playlist_name = 'Morning Energy'
WHERE playlist_id = 101;

SELECT *
FROM playlist
WHERE playlist_id = 101;
```